

Detailed Output Analysis of Speech Emotion Recognition Models

1 Introduction

When documenting Speech Emotion Recognition (SER) systems, understanding the exact data format and output types of each model is essential for downstream processing. This document elaborates on the precise structure, labels, and behavior of the outputs generated by SenseVoice, Whisper SER, and Wav2Vec 2.0.

2 SenseVoice (FunAudioLLM)

SenseVoice provides an in-context, rich text transcription mechanism. Instead of outputting standard probability matrices for an entire audio track, it embeds emotional metrics directly into the chronological stream of text.

2.1 Output Structure

- **Format:** A unified text string containing interleaved linguistic tokens (words) and non-linguistic markers (tags).
- **Core Emotional Labels:** [Happy], [Angry], [Sad], or implicit Neutral.
- **Acoustic Event Markers:** [Laughter], [Cry], [Cough], [Applause].

2.2 Sarcasm Profiling

Sarcasm is inferred via programmatic evaluation of text sentiment against the local emotional tag. For instance, if the transcript reads:

"Oh, fantastic [Angry]"

The semantic positivity of "fantastic" directly conflicts with the acoustic classification of "Angry", providing an empirical trigger for sarcasm detection.

3 Fine-Tuned Whisper (SER Variant)

Community fine-tuned variants of OpenAI's Whisper model strip away or bypass the text generation head to utilize the encoder's dense acoustic hidden states for multi-class emotion categorization.

3.1 Output Structure

- **Format:** A discrete probability distribution or array of floating-point confidence scores mapping to a fixed label set.
- **Categorical Labels:** The standard categorical emotion framework, typically consisting of: **Anger**, **Disgust**, **Fear**, **Happiness**, **Sadness**, **Surprise**, and **Neutral**.

3.2 Sarcasm Profiling

Because the model outputs a solitary macro-label or strict tensor distribution per audio snippet, sarcasm detection requires a dual-model pipeline: running a base Whisper instance to pull the semantic text, running the SER pipeline to pull the acoustic frame, and evaluating alignment via an exterior logic block.

4 Wav2Vec 2.0 (IEMOCAP Variant)

Models utilizing the Wav2Vec 2.0 architecture fine-tuned on the Interactive Emotional Dyadic Motion Capture (IEMOCAP) dataset are tuned to real human dialogue quirks rather than artificial acting scripts.

4.1 Output Structure

- **Format:** Floating-point confidence scores or log-odds per classified window.
- **Categorical Labels:** Restricted tightly to operational conversational dynamics: **Angry**, **Happy/Excited**, **Sad**, **Neutral**, and crucially, **Frustration**.

4.2 Sarcasm Profiling

The inclusion of the **Frustration** metrics acts as a reliable metric for sarcasm. In conversational linguistics, passive-aggressive remarks and sarcastic dismissals frequently trigger the acoustic properties of lower-register vocal tension, which map directly to the IEMOCAP **Frustration** baseline.

5 Summary Reference Table

Table 1: Model Output Characteristics

Model Framework	Primary Output Mechanism	Key Advantage for Sarcasm
SenseVoice	Inline Text/Event Markers	Seamless alignment with text context
Whisper SER	Categorical Probabilities	High acoustic structural granularity
Wav2Vec 2.0	Macro-State Logic	Dedicated Frustration taxonomy